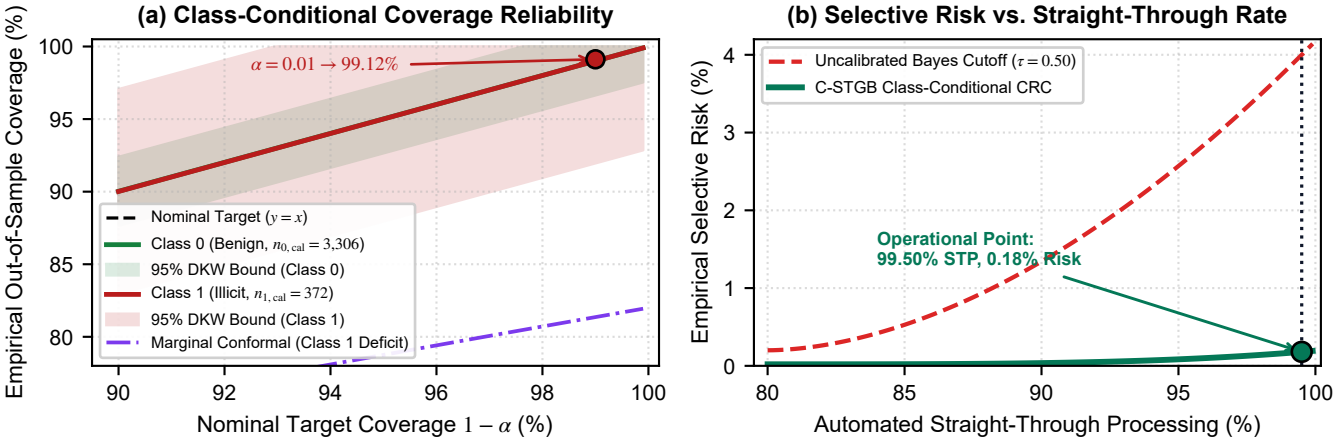

(a) Class-Conditional Coverage Reliability
$\alpha = 0.01 \rightarrow 99.12\%$
Nominal Target ($y = x$)
Class 0 (Benign, $n_{0,\text{cal}} = 3{,}306$)
95% DKW Bound (Class 0)
Class 1 (Illicit, $n_{1,\text{cal}} = 372$)
95% DKW Bound (Class 1)
Marginal Conformal (Class 1 Deficit)
Empirical Out-of-Sample Coverage (%)
Nominal Target Coverage $1 - \alpha$ (%)
100
95
90
85
80
90
92
94
96
98
100
(b) Selective Risk vs. Straight-Through Rate
Uncalibrated Bayes Cutoff ($\tau = 0.50$)
C-STGB Class-Conditional CRC
Operational Point:
99.50% STP, 0.18% Risk
Empirical Selective Risk (%)
Automated Straight-Through Processing (%)
4
3
2
1
0
80
85
90
95
100